

M&A OPERATING SYSTEM

Product Design

AI Analytics Platform

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About This Document

This document is part of the **M&A Operating System** product design specification series. It describes the intended design, architecture, and behavior of the platform within. It is intended for internal distribution across product, engineering, and design review functions.

This is a living specification. It reflects design intent at the time of publication and will be updated as the product evolves.

About M&A Operating System

M&A Operating System is a boutique advisory firm focused on helping organizations modernize the operational foundations required for scalable analytics, trusted enterprise information, and practical AI adoption.

Our work sits at the intersection of enterprise data strategy, information architecture, operating model modernization, and transformation execution. We specialize in helping organizations design practical, business-aligned approaches to enterprise information management that support operational scalability, analytics enablement, governance, and modern AI capabilities.

A core part of our approach is the recognition that successful data and AI initiatives are not driven solely by technology platforms. Long-term success depends on how well organizations structure, model, govern, operationalize, and align information with real business processes and decision-making.

Our Product Design document series reflects this philosophy. These documents are intended to provide practical frameworks, methodologies, architectural concepts, and implementation guidance that help organizations move beyond theoretical strategy into executable operational design.

M&A Operating System's approach combines: - Enterprise data and AI strategy - Subject-based data modeling and information architecture - Semantic and operational data design - Governance and organizational alignment - Operating model modernization - Transformation execution and delivery leadership - Practical AI operationalization and readiness

Our goal is simple: help organizations build practical, scalable, and trusted information foundations that improve operational effectiveness, accelerate analytics and AI adoption, and support better business outcomes.

Find Out More

To learn more about M&A Operating System, explore the platform, or speak with our team: maoperatingsystem.com

1. Executive Overview: Governed AI-Enabled Analytics and Data Mining

Introduction

About This Document

This technical white paper proposes an alternative approach to AI-enabled analytics and data mining in highly regulated and governed environments, where accuracy, traceability, and repeatability are critical. It is written for executives, architects, analytics leaders, and governance teams evaluating how to extend AI access to enterprise data.

The proposal combines AI interpretation with approved semantic definitions, controlled computation, and recorded evidence. The remaining sections explain the architecture, illustrate a technology stack, propose evaluation measures, examine Text-to-SQL risks, and discuss possible extensions. Examples are illustrative; this paper does not report deployment results or establish regulatory compliance.

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The Proposed Approach

We propose a governed semantic execution layer that gives people and AI applications access to approved calculations and datasets.

Business users need answers expressed in business terms: portfolio performance, risk exposure, liquidity, or operating outcomes. Those answers depend on agreed definitions, reliable source data, and permission to use it. When teams reconstruct those decisions for each report or request, they repeat work and risk producing different answers to the same business question.

The paper uses the name AI Analytics Platform for this proposed architecture. It makes approved analytical definitions reusable across conversations, applications, and automated analysis. A user asks a question through an existing interface. The platform identifies the approved calculation, checks access, executes it against registered sources, and returns the result with an explanation of how it was produced.

The platform sits above existing warehouses, data services, and specialist analytical engines. It requires supported connections and accurate mappings to those sources. Organizations can retain their existing investments while making governed analytics available to more consumers.

The proposed scope includes metric queries, drilldown, and governed bulk dataset retrieval for data-mining workflows. Responses include results, governed chart or table specifications, optional narrative summaries, and audit evidence. Consumer applications provide the user experience. The platform governs the dataset supplied to a data-mining workflow; it does not, by itself, validate downstream feature engineering, model training, or research conclusions. Section 2 demonstrates this boundary with a governed data-mining example.

Accuracy, Traceability, and Repeatability

The proposal addresses three distinct requirements. Each needs its own evidence before adoption.

Requirement	Meaning and Proposed Basis
Accuracy	A result answers the intended question using the correct business definition and suitable source data. Approved formulas, request validation, and comparison with independently verified results support this requirement. Consistent execution alone does not establish correctness.
Traceability	An authorized reviewer can follow a result back to the request, definition versions, permissions, source references, and execution decisions. The proposed lineage record preserves those links under an agreed retention policy.
Repeatability	An authorized rerun using the same request, data snapshot, definitions, permissions, and calculation settings produces the same result within a declared numerical tolerance. Version and snapshot preservation make this test possible.

For data mining, these requirements apply to the extracted dataset as well as to calculated metrics: which records and fields were supplied, under which selection rules, and from which source state. Subsequent model training or analysis requires its own validation.

Why Text-to-SQL Is Not the Long-Term Foundation

We do not believe Text-to-SQL alone is a sustainable long-term foundation for AI-enabled analytics in a highly regulated organization.

Text-to-SQL uses a large language model (LLM) to turn a question into a database query. These implementations are easy to stand up and quick to demonstrate: a model receives database schemas, generates a query, and returns an answer through a conversational interface. This makes them an attractive starting point for AI-enabled analytics and exploration.

The challenge is moving from a successful demonstration to recurring business metrics, data-mining workflows, automated decision inputs, and regulatory submissions. A convincing answer in a demonstration does not show that the calculation is correct, that a reviewer can trace it, or that an authorized rerun can repeat it. This paper proposes governed semantic execution as the alternative foundation for those workloads.

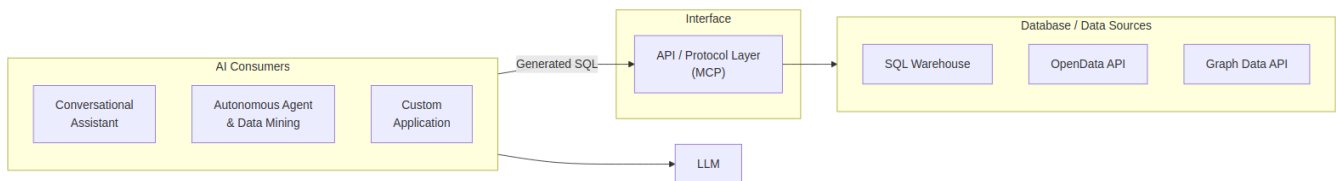
A generated query does not, by itself, establish which business definition was approved, which version applied, or whether the result can be reproduced. Database permissions and query validation can restrict execution, but they do not decide what a regulated metric means. Organizations still need accountable owners, approved formulas, controlled changes, and evidence linking each result to those decisions.

The proposed platform makes those requirements part of the execution process. AI selects from approved analytical definitions; deterministic software constructs and executes the calculation. This reduces reliance on the model inferring business meaning from physical schemas on each request.

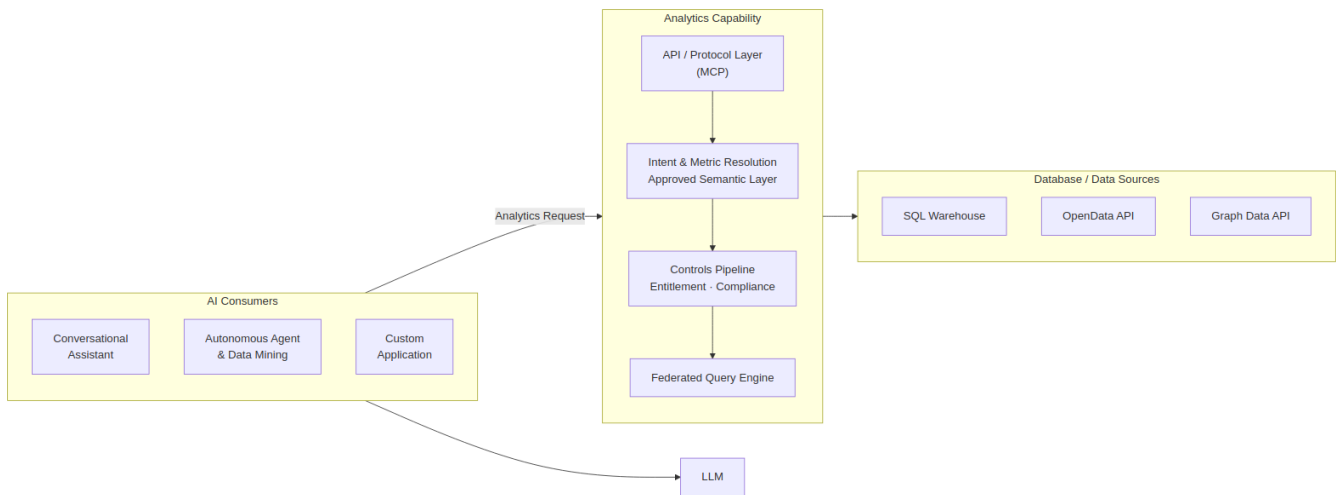
Proposed Architecture at a Glance

In both approaches, AI clients and LLMs are present. The difference is what happens next. In Text-to-SQL, the AI generates queries directly against raw database schemas. In the AI Analytics Platform, it submits structured requests to a governed semantic layer, and the platform handles all computation, governance, and execution.

Text-to-SQL Approach



AI Analytics Platform



In the platform approach, every request routes through the API layer, traverses the invariant controls sequence, and produces an audit record. No path to execution backends, physical schemas, or raw data exists outside that pipeline.

The Text-to-SQL appendix explains the appropriate role of Text-to-SQL and the boundary between controlled exploration and governed execution.

Text-to-SQL remains useful alongside the platform for controlled exploration and discovering new analytical needs. A shared conversational interface can support both paths, provided users can distinguish exploratory output from governed results. Useful exploratory calculations can become candidates for review and registration. AI may help draft those definitions; accountable people approve them before governed use.

How the Platform Works

In a governed AI-enabled analytics situation, the Semantic Metrics Repository (SMR) provides a catalog of approved metrics, dimensions, operations, and dataset contracts. Each definition states what the analysis means, how it is calculated, and which data it uses. Versioning preserves the definition associated with a historical result.

A request follows a common sequence:

1. **Understand the question.** AI matches a natural-language request to registered operations and asks for clarification when needed. Applications that already know the required operation can submit it directly.
2. **Check permission and controls.** The platform checks identity, permitted metrics and data, query size, complexity, and compliance purpose before execution.
3. **Calculate the result.** The platform builds a query from approved definitions and executes it across supported sources. AI does not write the executable query or calculate the returned values.
4. **Present and record the answer.** The platform returns data with a governed presentation specification and an optional AI summary. It records the definitions, access decisions, and execution details needed to explain the result.

AI consumers connect through the Model Context Protocol (MCP), a standard interface for tools and data. The same controls apply to people using conversational assistants and to autonomous applications.

For a compliance-relevant metric requested for a compliance purpose, the design adds a signed evidence package and an export gate. Export remains blocked until the required evidence is sealed. This supports review; it does not replace the organization's approval process or establish regulatory compliance by itself.

Reproducibility depends on preserving the relevant data snapshot, definition versions, access policies, and calculation settings. A consistent formula cannot produce an identical historical result from changed source data.

Examples by Persona

User Need	Platform Response
A portfolio manager compares returns with a benchmark.	The platform applies the approved return definitions to authorized portfolios and returns a comparison with its calculation record.
A risk officer investigates a limit breach.	The platform combines registered risk and limit data, identifies breaches, and supports further investigation within the officer's permitted scope.
A research application retrieves historical positions.	The platform supplies an approved dataset, limits fields and rows to the application's permissions, and records the retrieval.
A treasury analyst prepares liquidity figures for a submission.	The platform calculates the approved liquidity coverage ratio and produces the additional evidence required by its compliance-purpose controls.

These scenarios use the same governed service. Their differences are the approved operation, the caller's permissions, the requested output, and the evidence required.

Supporting Roles and Activities

The platform depends on accountable ownership of definitions, permissions, and source data.

Data Modellers describe the source data and its business meaning. Metrics Modellers define calculations and dataset contracts. Analytics Governance approves definitions and oversees analytical quality. The Entitlements Manager maintains access policies, Integration Engineers connect sources, and Platform Admins operate the service.

This work determines what the platform can answer. An unregistered business concept requires a reviewed definition before it becomes available through the governed path. Catalog coverage therefore grows through a maintained approval process.

The platform cannot correct an unsuitable formula or poor source data merely by executing consistently. Owners must establish data quality, freshness, retention, and acceptable uses. Security and compliance teams must validate the controls for the intended deployment.

Design Principles

The following principles retain the identifiers used throughout the specification.

Principle	Executive Meaning
P1 - Semantic abstraction	Users work with approved business concepts. Access to physical execution details is separately controlled.
P2 - Controls before execution	Requests must pass the required checks before data-source execution begins.
P3 - Deterministic metric resolution	Each metric resolves to an approved, versioned definition.
P4 - Complete analytical lineage	Accepted requests produce evidence of the definitions, decisions, and execution or failure involved.
P5 - Role-aware by default	Access requires explicit permission and is restricted to the caller's authorized data.
P6 - Governed narrative	AI summaries are checked against computed results and withheld when validation fails.
P7 - Deterministic visualization	Registered rules select the chart or table format for the result.
P8 - Explainability at every layer	Authorized users can inspect the interpretation, calculation, and supporting evidence.
P9 - Administrator sovereignty within governance bounds	Administrators configure the service within mandatory governance constraints.
P10 - Deterministic computation, not generation	Approved software computes values from registered definitions; AI interprets requests and summarizes results.

Implementation, Success, and Development

The architecture separates required behavior from technology choices. The proposed reference stack uses Python services, Starburst for federated query execution, PostgreSQL and object storage for policies and evidence, Redis for caching and concurrency, and language models for interpretation and summaries. These choices require validation against the organization's sources, scale, and operating requirements.

Success means that users obtain the analysis they need with approved definitions, appropriate access, and usable evidence. The organization should measure adoption, definition coverage, data freshness, response time, failures, cost, and the integrity of the audit record. Proposed targets require workload testing and a baseline; they are not demonstrated results.

Possible future extensions include scheduled monitoring, alerts, saved queries, and collaboration. They are outside the proposed governed core and are not commitments about current availability or delivery dates.

Supporting Section	What It Adds
2. Proposed Architecture and Capabilities	Defines roles, component responsibilities, interfaces, and controls.
3. Glossary	Defines the named components and technical terms.
Appendix A. Text-to-SQL	Defines the boundary between controlled exploration and governed execution.
Appendix B. Illustrative Reference Implementation and Evaluation Plan	Applies the logical design to one proposed technology stack and defines how that implementation should be evaluated.

The Executive Decision

Adoption requires commitment to a governed analytical service and the people who maintain its definitions and controls.

A bounded evaluation should establish whether the proposed platform can serve a meaningful set of business questions across the required sources, enforce existing permissions, and produce evidence that reviewers can use. Leadership should name the accountable owners and agree the measures of success before that evaluation begins.

The decision to expand should follow evidence that users can obtain reliable analysis with less repeated definition and reconciliation work, while the organization retains control over access, calculation changes, and audit records.

2. Proposed Architecture and Capabilities

Purpose and Boundaries

This section defines the proposed logical architecture for governed AI-enabled analytics and data mining. It explains each component's responsibility and the contracts between components. It is implementation-neutral: it names required behavior, inputs, outputs, and evidence without selecting products, programming languages, frameworks, storage services, or deployment patterns. Appendix B maps these responsibilities to one illustrative technology stack.

The design separates AI-assisted interpretation from governed computation. A language model may match a question to an approved operation and summarize a completed result. Registered definitions, deterministic software, and controlled data services perform the calculation. A structured caller may bypass natural-language interpretation, but it may not bypass entitlements, validation, controls, execution evidence, or other mandatory stages.

Component names identify logical responsibilities. They do not require one independently deployed service per component. Required behaviors describe the proposal, not verified features of a released product.

Platform Roles

The operating model separates analytical use, definition ownership, access policy, governance, integration, and platform operation. One person may hold several roles, but the organization must review combinations that create conflicts of interest.

Role	Primary Responsibility
Analytical End User	Requests governed analysis and views results within assigned permissions.
Power Analyst	Performs deeper exploration, drilldown, and authorized evidence export.
Data Modeller	Maintains business and physical data metadata in the Semantic Data Repository (SDR).
Metrics Modeller	Authors metric, dimension, operation, and dataset definitions for review.
Entitlements Manager	Maintains analytical access, row-scope, and masking policies.
Analytics Governance	Approves definitions and oversees analytical quality and catalog integrity.
Integration Engineer	Registers supported source connections and physical mappings.
Platform Admin	Operates the service within governance policy. This role does not confer analytical data access.

Production access must come from approved entitlement policy, not from an authoring or administrative role alone.

Role and Capability Boundaries

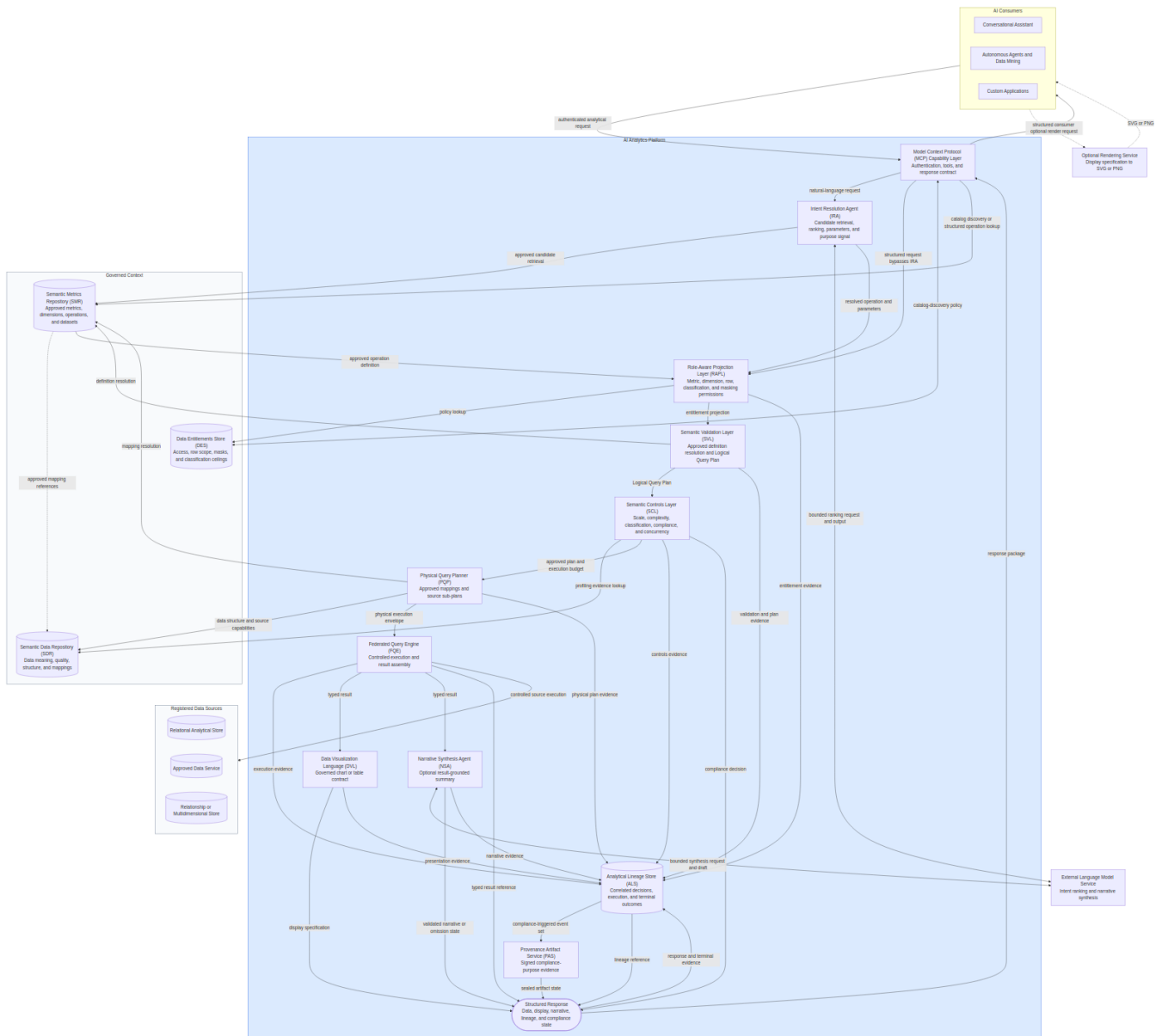
Activity	Accountable Role	Required Separation
Use approved analytics	Analytical End User or Power Analyst	Access remains limited by Data Entitlements Store (DES) policy.
Author data metadata and mappings	Data Modeller	Production approval remains with the designated governance process.
Author metrics and operations	Metrics Modeller	The author cannot make a definition available merely by submitting it.
Approve analytical definitions	Analytics Governance	Approval is recorded against a versioned definition.
Maintain entitlement policy	Entitlements Manager	Platform administration does not imply authority to grant data access.

Activity	Accountable Role	Required Separation
Register source connections	Integration Engineer	Credentials remain inside the controlled execution boundary.
Operate platform services	Platform Admin	Administrative access does not imply permission to run analytical queries.

The implementation must record who proposed, reviewed, approved, deprecated, and used each governed definition. Organizations may assign these responsibilities differently, but they must preserve accountable approval and prevent infrastructure privileges from becoming analytical entitlements.

Architecture and Request Flow

The governed path applies the same controls to conversational users, autonomous agents, and custom applications.



Component Interaction Map

The architecture is not a strictly linear sequence. Some components provide lookups, several write evidence during the request, structured requests bypass IRA, and presentation components run in parallel. The component contracts below therefore associate inputs and outputs with the component that owns them rather than with a numbered step.

Component	Business Role	Receives	Provides
AI Consumers	Start governed analysis and present the answer.	Business question or approved operation; structured response	Authenticated analytical request; optional render request
MCP Capability Layer	Govern the platform boundary and public contract.	Authenticated request; DES catalog-discovery policy; assembled response package	Entitlement-scoped catalog view; correlated natural-language request to IRA; structured request with approved SMR operation reference to RAPL; structured response to consumer
Intent Resolution Agent (IRA)	Translate business language into an approved analytical request.	Natural-language request; approved candidates from SMR; bounded model output	Resolved operation and parameters; confidence and purpose evidence; clarification request when needed
Semantic Metrics Repository (SMR)	Govern reusable analytical meaning and versions.	Authoring changes; discovery and definition lookups; mapping lookups	Approved operations, metrics, dimensions, datasets, presentation metadata, and mapping references
Semantic Data Repository (SDR)	Govern data meaning, structure, quality, and mapping context.	Approved data metadata and lineage updates	Versioned data context and mapping targets referenced by SMR and PQP
Data Entitlements Store (DES)	Govern business-level analytical access policy independently.	Approved role and access-policy changes	Versioned metric, dimension, row-scope, mask, and classification policies
Role-Aware Projection Layer (RAPL)	Determine the caller's permitted analytical scope.	Resolved request; authenticated identity; DES policies	Entitlement projection to SVL; entitlement evidence to ALS
Semantic Validation Layer (SVL)	Turn an entitled request into a valid logical calculation.	Resolved request; entitlement projection; approved SMR definitions	Logical Query Plan to SCL; validation and plan evidence to ALS; structured rejection
Semantic Controls Layer (SCL)	Decide whether a valid plan may execute under current policy and conditions.	Logical Query Plan; versioned controls; profiling evidence; operating state	Approved plan and budget to PQP; controls evidence to ALS; compliance decision to response assembly; structured rejection
Physical Query Planner (PQP)	Translate logical calculations into bounded source instructions.	Approved Logical Query Plan; execution budget; SMR and SDR mapping context	Physical execution envelope to FQE; plan integrity evidence to ALS

Component	Business Role	Receives	Provides
Federated Query Engine (FQE)	Execute approved source instructions and assemble controlled results.	Physical execution envelope; registered source connections	Typed result to DVL, NSA, and response assembly; execution evidence to ALS; explicit terminal outcome
Registered Data Sources	Execute bounded operations over governed business data.	Authorized source instructions from FQE	Typed source data or explicit failure to FQE
Data Visualization Language (DVL)	Define a consistent governed presentation.	Typed result; resolved intent; approved labels and units	Display specification to response assembly; presentation evidence to ALS
Narrative Synthesis Agent (NSA)	Produce an optional result-grounded explanation.	Bounded result content; bounded model draft	Validated narrative or omission state to response assembly; narrative evidence to ALS
External Language Model Service	Support bounded language ranking and drafting.	Candidate-ranking request from IRA; synthesis request from NSA	Ranking or draft text back to the requesting component
Analytical Lineage Store (ALS)	Preserve the correlated evidence chain across the request.	Intent, entitlement, validation, controls, plan, execution, and presentation events	Lineage reference to response assembly; event set to PAS and audit export
Provenance Artifact Service (PAS)	Seal additional evidence for compliance-purpose results.	Active compliance trigger; correlated ALS event set	Sealed artifact-state reference or failure state to response assembly; export state
Structured Response	Assemble the governed delivery package.	Typed result; display; narrative; lineage; compliance decision; optional artifact state; warnings and errors	Response package to MCP Capability Layer; response and terminal evidence to ALS
Optional Rendering Service	Produce a static visual without joining analytical computation.	Self-contained display specification from a consumer	SVG or PNG to the consumer

The running example uses one request throughout. Each primary flow component shows both its input and output as technology-neutral JSON. When a downstream contract requires a request, definition, projection, plan, result, or evidence reference, it reuses the same field name and value so the handoff is visible. Identifiers and values are illustrative, not implementation defaults.

A result is repeatable only when the request, source snapshot, definition versions, effective permissions, configuration, and execution software are preserved. Deterministic planning does not compensate for changed data or definitions.

AI Consumers

Business definition. AI consumers are the channels through which people and automated processes request and use governed analysis.

Why this role exists. The architecture separates the user experience from analytical governance so that several interfaces can use the same approved definitions, permissions, controls, and evidence.

Input and output. A consumer starts with a business question or a known approved operation and the caller's authenticated context. It outputs a governed analytical request and later renders the structured response without recalculating or reinterpreting its values.

AI consumers include conversational assistants, autonomous agents, data-mining workflows, and custom applications. They provide the user experience; they do not own analytical definitions, permissions, query planning, or execution.

A consumer may submit natural language or an approved operation identifier with typed parameters. It passes the caller's authenticated context, preserves the returned result identifier for follow-up actions, and distinguishes governed results from exploratory output produced outside this platform.

Input and Output Example

The running example asks the platform to compare the caller's equity portfolios with their benchmarks for the current quarter:

```
{
  "input": {
    "business_request": "Compare my equity portfolios with their benchmarks this quarter"
  },
  "output": {
    "analytical_request_id": "request-draft-001",
    "question": "Compare my equity portfolios with their benchmarks this quarter"
  }
}
```

The authentication and identity context travels through the trusted transport. It is not embedded in the analytical request body.

The consumer request becomes the MCP Capability Layer input, beginning the governed pipeline.

MCP Capability Layer (MCP)

Governing principles: P1 - Semantic abstraction, P2 - Controls before execution, and P5 - Role-aware by default

Business definition. The MCP Capability Layer is the governed front door to the analytical service.

Why it exists. It gives every consumer one consistent contract and prevents conversational, automated, or custom interfaces from creating alternative paths around authentication and governance.

Input and output. It receives an authenticated natural-language or structured request. It outputs a correlated request to the correct pipeline stage and, after processing, a structured response to the consumer.

The MCP Capability Layer is the single governed analytical entry point. The transport authenticates the caller before tool routing and passes identity through trusted request context. Bearer tokens are not analytical tool arguments. A structured request bypasses only language interpretation: the supplied operation identifier is resolved to an approved SMR definition before the request reaches RAPL.

Tool Catalog

Tool	Purpose
run_analytics	Accepts either a natural-language question or an approved operation with typed parameters. Natural language is resolved by the IRA before governed execution begins.
list_operations	Lists operations and parameters available to the caller.
drilldown	Creates a governed follow-up request from a prior result and hierarchy selection.

A drilldown inherits analytical context but receives a new request identifier and passes the current entitlement and controls pipeline again. Prior approval never authorizes a later request automatically.

Execution Profiles

Profile	Result
data_retrieval	Returns a typed, paginated dataset governed by an approved dataset contract.
metric_query	Returns calculated data and a DVL display specification.
full_analytical	Adds an optional governed narrative to the metric response.

Execution profiles control optional response assembly, not governance. Every profile uses the complete entitlement, validation, controls, planning, execution, and lineage path. PAS invocation is compliance-driven and independent of presentation profile.

Intent Confirmation

When the IRA cannot select one operation confidently, the MCP layer returns ranked confirmation cards containing the proposed operation, bound parameters, material filters, and presentation preview. Refinement does not execute a backend query. The selected request still passes all downstream controls.

Confidence thresholds, candidate counts, and refinement limits require evaluation. They are not accuracy guarantees.

A confirmation card contains the candidate operation, resolved parameter values, time period, material filters, confidence, and a presentation preview. It must not expose physical schemas or executable queries. The session binds the candidate set to the authenticated caller and expires after a short configured interval.

The consumer may select a candidate, refine the question, or cancel. Selection resumes the original governed request; refinement produces a new ranked candidate set without querying a backend. Operations marked as requiring confirmation always present a card, regardless of confidence.

Capability Governance

The capability layer exposes only registered tools and approved operation metadata. Tool and catalog discovery apply a bounded DES policy view before IRA candidate retrieval. This view limits which metadata the caller may discover; it does not authorize execution. RAPL independently evaluates the complete request and current entitlement policy before validation or planning. Request limits, session limits, and allowed response sizes are versioned platform controls. Changes to tool contracts follow the same review and compatibility process as other public interfaces.

Input and Output Example

The MCP layer accepts the natural-language request and creates a correlation identifier before routing it to the IRA:

```
{
  "input": {
    "authenticated_context_ref": "caller-context",
    "analytical_request_id": "request-draft-001",
    "question": "Compare my equity portfolios with their benchmarks this quarter"
  },
  "output": {
    "request_id": "req-20260518-093241",
    "route": "natural_language",
    "analytical_request_id": "request-draft-001",
    "question": "Compare my equity portfolios with their benchmarks this quarter",
    "discoverable_catalog_ref": "approved-operations-for-caller"
  }
}
```

For natural-language requests, the correlated request becomes the IRA input. A structured request bypasses IRA, resolves its operation identifier against the SMR, and then proceeds to RAPL with the approved operation reference.

Intent Resolution Agent (IRA)

Governing principles: P1 - Semantic abstraction and P10 - Deterministic computation, not generation

Business definition. The IRA translates a user's business language into a registered analytical operation and its typed parameters.

Why it exists. Users should not need to know catalog identifiers, but the platform must not let a language model invent calculations or executable queries. The IRA bridges that gap and asks for confirmation when meaning is uncertain.

Input and output. It receives a natural-language request and the approved definitions discoverable by the caller. It outputs a resolved operation, parameters, confidence evidence, and purpose signal, or a clarification request.

The IRA is the only AI step before computation. It retrieves candidate operations from the SMR, ranks them against the user's request, binds parameters, and returns either a resolved request or a confirmation choice. It does not receive credentials, execute queries, or generate executable SQL.

The IRA uses registered descriptions rather than physical schemas. Caller-dependent phrases such as "my portfolios" remain unresolved until RAPL evaluates the authenticated identity. The output contains an approved operation identifier, typed parameters, candidate evidence, model details, and a confidence score.

The IRA also estimates whether the stated purpose is compliance-related. Ambiguous purpose requires clarification. The score is an input to a deterministic control; the model does not make the final control decision. Structured callers bypass the IRA and declare purpose through a validated field.

Resolution Pipeline

Stage	Required Behavior	Evidence Retained
Candidate retrieval	Search approved SMR operation and metric descriptions within the caller's discoverable scope.	Query embedding version, retrieved identifiers, and retrieval scores.
Candidate ranking	Rank candidates and bind parameters using the natural-language request and registered descriptions.	Model and prompt-template versions, ranked identifiers, confidence, and bound parameters.
Ambiguity check	Compare the leading candidates and apply operation-specific confirmation rules.	Thresholds, score difference, and confirmation decision.
Purpose classification	Estimate whether the stated purpose is compliance-related.	Purpose score and any clarification outcome.
Output validation	Validate the selected operation identifier and typed parameters against the SMR contract.	Validation result and rejection reason when applicable.

Retrieval narrows the model's choice set; it does not approve a definition or establish that the chosen intent is correct. Evaluation must measure top-candidate accuracy, correct-candidate coverage, clarification quality, and false certainty across representative language and user roles.

Input and Output Example

The IRA resolves the question to an approved comparison operation. The symbolic caller-dependent scope remains unresolved:

```
{
  "input": {
    "request_id": "req-20260518-093241",
    "question": "Compare my equity portfolios with their benchmarks this quarter",
    "discoverable_catalog_ref": "approved-operations-for-caller"
  },
  "output": {
    "resolved_request_id": "resolved-20260518-093242",
    "request_id": "req-20260518-093241",
    "operation_id": "compare_portfolio_to_benchmark",
    "parameters": {
      "portfolio_scope": "caller_authorized_portfolios",
      "asset_class": "EQUITY",
      "time_period": "current_quarter"
    },
    "confidence": 0.93,
    "compliance_purpose_score": 0.08,
    "confirmation_required": false
  }
}
```

The IRA uses the SMR to limit interpretation to approved analytical choices. Its resolved operation then becomes part of the RAPL input.

Semantic Metrics Repository (SMR)

Governing principles: P1 - Semantic abstraction, P3 - Deterministic metric resolution, and P10 - Deterministic computation, not generation

Business definition. The SMR is the organization's governed rulebook for analytical meaning: what a metric, dimension, operation, or dataset means and which approved version applies.

Why it exists. Without a shared catalog, each report, application, or AI interaction can reconstruct the same business concept differently. The SMR makes approved definitions reusable and versioned.

Input and output. Authors submit proposed definitions through governance. At runtime, the IRA, SVL, and PQP submit approved identifiers or discovery queries. The SMR outputs discoverable operation metadata, pinned semantic definitions, and approved mapping references.

In a governed AI-enabled analytics situation, the Semantic Metrics Repository (SMR) provides a catalog of approved metrics, dimensions, operations, and dataset contracts. It is the authoritative source for concepts that the governed path can resolve.

Definition Type	Required Content
Metric	Formula or governed measure, aggregation, units, dimensions, classification, source affinity, owner, version, and approval state.
Dimension	Business meaning, hierarchy, compatible metrics, and mapping information.
Operation	Parameter contract, supported metrics and dimensions, execution profile, and confirmation requirements.
Dataset	Approved fields, classifications, selection rules, mapping, refresh expectations, and pagination policy.

Only versions in the Approved state are normally resolvable for new governed requests. A Deprecated version may remain temporarily resolvable only when an explicit transition policy permits it. Historical versions remain available to authorized reviewers for lineage reconstruction.

Definition Lifecycle and Versioning

State	Meaning
Proposed	An author has submitted a draft that cannot be used for governed execution.
In Review	Named reviewers are assessing meaning, sources, tests, ownership, and access implications.
Approved	The exact version may be resolved by governed requests.

State	Meaning
Deprecated	Existing evidence remains interpretable, but new use is discouraged or blocked according to policy.
Retired	The version is unavailable for new execution and retained only for authorized historical reconstruction.

Approving a new version does not rewrite earlier lineage. Each request pins the versions it resolved. Compatibility rules determine whether consumers must change when parameters, dimensions, units, formulas, mappings, or classifications change.

Minimum Definition Contract

Every definition carries a stable identifier, semantic version, owner, approval state, effective period, classification, and change history. Metrics additionally declare their formula or governed measure, aggregation behavior, units, compatible dimensions, source affinity, physical mapping reference, and tests. Operations declare typed parameters, output expectations, execution profile, permitted drilldowns, and whether confirmation is mandatory. Dataset contracts declare approved fields, classifications, selection rules, ordering, pagination, and freshness expectations.

The SMR and Semantic Data Repository (SDR) are separate stores within the Data Context Store (DCS). The SMR defines analytical meaning; the SDR describes organizational data and its physical structure. Approved mappings link the two without exposing physical schema details through the normal AI interface.

Formula Language

The formula language expresses metric logic using registered metric identifiers, logical fields, arithmetic, conditions, safe division, time windows, filtered aggregation, and other explicitly supported operations. It does not accept arbitrary SQL or backend-specific expressions.

A formula is human-readable and audit-visible, but registration does not prove correctness. Approval requires accountable ownership, test cases, source validation, and comparison with independently verified results.

Authoring and Discovery

Metrics Modellers submit definitions through an authenticated workflow. Analytics Governance reviews and approves them. Consumers discover only approved operations available under their entitlements. Search indexes and embeddings may improve discovery; they do not change approval state or access rights.

Input and Output Example

The resolved operation references two approved metrics without exposing their physical implementation:

```

{
  "input": {
    "operation_id": "compare_portfolio_to_benchmark",
    "required_state": "approved"
  },
  "output": {
    "operation_definition_ref": "smr:operation:compare_portfolio_to_benchmark@1.3.0",
    "operation_id": "compare_portfolio_to_benchmark",
    "version": "1.3.0",
    "status": "approved",
    "metrics": [
      {
        "metric_id": "portfolio_return",
        "version": "2.1.0"
      },
      {
        "metric_id": "benchmark_return",
        "version": "1.4.2"
      }
    ],
    "dimensions": [
      "portfolio"
    ],
    "required_parameters": [
      "portfolio_scope",
      "asset_class",
      "time_period"
    ],
    "intent_pattern": "comparison",
    "presentation_metadata_ref": "smr:presentation:portfolio-benchmark-comparison@2.0",
    "execution_profile": "full_analytical"
  }
}

```

The approved operation definition and resolved request now move to RAPL, where the caller's permitted analytical scope is determined.

Role-Aware Projection Layer (RAPL)

Governing principle: P5 - Role-aware by default

Business definition. RAPL converts organizational access policy into the exact analytical scope permitted for the current caller and request.

Why it exists. Permission to reach a data service does not explain which business metrics, dimensions, populations, or fields a person may use. RAPL makes those business-level restrictions explicit before planning begins.

Input and output. It receives the resolved request, authenticated identity context, and versioned DES policies. It outputs an entitlement projection containing approved concepts, row scope, masks, classification ceiling, and decision evidence.

RAPL converts authenticated identity claims and policies from the Data Entitlements Store (DES) into an effective entitlement projection. It validates identity context, retrieves role definitions, merges them according to policy, resolves claim-based row scopes, and records the decision.

Restriction	Proposed Treatment
Data and metric access	Deny any required domain, classification, or metric outside the effective access set.
Dimension access	Reject a required unauthorized dimension; do not silently change the question.
Row scope	Resolve typed predicates from trusted claims and attach them to the plan.
Column masking	Carry each required mask and its policy basis into planning and execution.
Classification ceiling	Preserve the effective ceiling for validation and controls.

The proposed multi-role rule uses union for granted data, metrics, and dimensions; strict intersection for row scopes; and union for masks, so any applicable mask remains in force. Explicit-deny and missing-claim behavior must be defined consistently in DES policy and tested independently of role order.

Row scopes are typed predicates, not interpolated query fragments. Masks are enforced before protected values leave the controlled execution boundary. Each projection records the roles, policy versions, resolved scopes, masks, and decision basis in the ALS.

Multi-Role Merge and Deny Behavior

Entitlement	Proposed Merge Rule
Data domains, metrics, and dimensions	Union of grants, subject to explicit-deny policy and classification ceiling.

Entitlement	Proposed Merge Rule
Row scopes	Strict intersection. Every applicable population restriction remains in force.
Column masks	Union. A mask required by any applicable role remains active.
Classification ceiling	Most restrictive applicable ceiling unless policy explicitly defines another reviewed rule.

The merge must be independent of role order and must fail closed when a required role definition or claim cannot be resolved. The design decision about whether an explicit deny overrides every grant must be uniform across DES, RAPL, tests, and audit explanations.

Column Masking Modes

Mode	Result Treatment
null replacement	Replace the protected value with null while preserving the field.
redacted label	Replace the value with a fixed approved label.
excluded	Remove the field from the returned schema.
hash replacement	Replace the value with a deterministic one-way representation when approved grouping behavior is required.

Masking changes what downstream consumers can observe. The selected mode, protected field, policy version, and enforcement location therefore form part of the analytical evidence.

Input and Output Example

RAPL resolves the symbolic portfolio scope from the authenticated caller's policy and claims:

```

{
  "input": {
    "resolved_request_id": "resolved-20260518-093242",
    "request_id": "req-20260518-093241",
    "operation_definition_ref": "smr:operation:compare_portfolio_to_benchmark@1.3.0",
    "authenticated_context_ref": "caller-context",
    "entitlement_policy_ref": "current-policy-snapshot"
  },
  "output": {
    "entitlement_projection_ref": "projection:req-20260518-093241",
    "request_id": "req-20260518-093241",
    "policy_version": "portfolio-access-7.2",
    "approved_metrics": [
      "portfolio_return",
      "benchmark_return"
    ],
    "approved_dimensions": [
      "portfolio"
    ],
    "row_scope": {
      "field": "portfolio_id",
      "operator": "in",
      "values": [
        "GLOB_EQ_OPP",
        "UK_CORE_INC",
        "ASIA_PAC_GRW",
        "EUR_BAL_INC"
      ]
    },
    "column_masks": [],
    "classification_ceiling": "INTERNAL",
    "decision": "approved"
  }
}

```

The entitlement projection becomes an SVL input alongside the resolved request and pinned SMR definitions.

Semantic Validation Layer (SVL)

Governing principles: P2 - Controls before execution and P10 - Deterministic computation, not generation

Business definition. The SVL is the semantic quality gate that turns an entitled business request into a precise, technology-neutral calculation plan.

Why it exists. A recognized intent can still be incomplete, incompatible, unapproved, or inconsistent with the caller's permissions. The SVL stops those requests before physical planning or source access.

Input and output. It receives the resolved request, entitlement projection, and pinned SMR definitions. It outputs a backend-independent Logical Query Plan or a structured rejection.

The SVL validates a fully qualified request and compiles it into a platform-independent LQP. No AI model runs in this component.

The SVL validates structure and parameter types; resolves operations, metrics, dimensions, and datasets to approved versions; checks semantic compatibility; applies the RAPL projection; attaches validated compliance inputs; and creates a directed plan containing logical concepts rather than SQL, connection details, or physical identifiers.

Unknown, unapproved, incompatible, or unauthorized required concepts cause a structured rejection. The SVL does not return a partial calculation that changes the requested meaning. Its validation shows conformity to registered definitions; it does not establish that the definitions or source data are correct.

Validation Stages

Stage	Checks	Output
Request validation	Required fields, types, parameter ranges, and operation contract.	A well-formed request or structured rejection.
Semantic resolution	Approval state, pinned versions, metric-dimension compatibility, and dataset contract.	Fully resolved logical identifiers.
Entitlement application	Permitted concepts, row predicates, masks, and classification ceiling.	An entitlement-constrained request.
Plan generation	Logical scans, filters, calculations, joins, sorts, limits, and output shape.	A backend-independent LQP.

The LQP also carries definition versions, data-affinity hints, mask directives, compliance inputs, and a preliminary impact indicator. It contains no credentials, connection endpoints, physical table names, or executable SQL.

Input and Output Example

The SVL converts the qualified request into a backend-independent plan:

```

{
  "input": {
    "resolved_request_id": "resolved-20260518-093242",
    "operation_definition_ref": "smr:operation:compare_portfolio_to_benchmark@1.3.0",
    "entitlement_projection_ref": "projection:req-20260518-093241"
  },
  "output": {
    "lqp_id": "lqp-20260518-093243",
    "request_id": "req-20260518-093241",
    "operation_id": "compare_portfolio_to_benchmark",
    "execution_profile": "full_analytical",
    "intent_pattern": "comparison",
    "presentation_metadata_ref": "smr:presentation:portfolio-benchmark-comparison@2.0",
    "entitlement_projection_ref": "projection:req-20260518-093241",
    "metric_versions": {
      "portfolio_return": "2.1.0",
      "benchmark_return": "1.4.2"
    }
  },
  "steps": [
    {
      "type": "metric_scan",
      "metrics": [
        "portfolio_return",
        "benchmark_return"
      ]
    },
    {
      "type": "filter",
      "field": "asset_class",
      "operator": "eq",
      "value": "EQUITY"
    },
    {
      "type": "row_scope",
      "field": "portfolio_id",
      "operator": "in",
      "value_ref": "projection:req-20260518-093241"
    },
    {
      "type": "sort",
      "field": "portfolio_return",
      "direction": "descending"
    }
  ],
  "time_period": {
    "type": "relative",
    "value": "current_quarter"
  },
  "compliance_inputs": {
    "metric_signal": false,
    "purpose_score": 0.08
  },
  "column_masks": []
}

```

The completed Logical Query Plan becomes the SCL input for the mandatory release decision.

Semantic Controls Layer (SCL)

Governing principles: P2 - Controls before execution, P8 - Explainability at every layer, and P9 - Administrator sovereignty within governance bounds

Business definition. The SCL is the operational and governance release gate for an otherwise valid analytical plan.

Why it exists. Entitlement and semantic validity do not address excessive scale, unsafe complexity, operating capacity, classification boundaries, or compliance-purpose handling. The SCL decides whether the plan may proceed under current policy and conditions.

Input and output. It receives the Logical Query Plan, versioned control policy, profiling evidence, and operating state. It outputs an approved plan with an execution budget or a structured rejection, together with a recorded decision.

The SCL is the mandatory release gate between logical validation and physical planning. It evaluates every LQP against versioned controls and records the decision before execution.

Control	Decision
Data scale	Compare an evidence-based scan estimate with the configured limit.
Complexity	Check plan nodes, join depth, source count, and other bounded measures.
Classification	Compare the highest required classification with the permitted ceiling.
Compliance	Evaluate approved metric metadata and declared or classified purpose.
Concurrency	Check active work against the operating limit.
Timeout	Assign a bounded execution budget after blocking checks pass.

Estimates are not exact measurements. The record identifies the source and freshness of the statistics used. Rejections return structured reasons and safe ways to narrow the request.

The proposed compliance trigger requires both a compliance-relevant metric and a compliance purpose. For natural-language requests, the SCL evaluates the IRA score against a configured threshold; structured requests declare purpose explicitly. The SCL records both signals. When both are active, it marks the provenance artifact as required and blocks export until PAS seals the completed evidence set after execution. Whether this rule is sufficient for a jurisdiction or activity remains a deployment-specific compliance decision.

Control Decision Contract

Each control records its configured limit, observed or estimated value, data source, evaluation version, outcome, and reason. The SCL evaluates all controls needed for a useful explanation, even when one failure is sufficient to block execution. Sensitive platform state, such as other tenants' workload details, is not returned to the caller.

The data-scale estimate uses current SDR profiling statistics and the resolved row scope. The complexity control considers structural cost separately from volume. The classification control applies both platform boundaries and the caller's effective ceiling. The concurrency control protects shared execution resources without weakening tenant isolation. After all blocking controls pass, the SCL assigns a timeout and writes the signed or integrity-protected decision record before releasing the plan.

No interactive user, automated agent, administrator, cache path, or internal retry bypasses this release decision. A retried or modified request receives its own decision record.

Input and Output Example

The SCL records each decision and assigns the execution budget:

```

{
  "input": {
    "lqp_id": "lqp-20260518-093243",
    "controls_policy_ref": "current-controls",
    "profiling_evidence_ref": "sdr-profile:portfolio-performance-source@2026-05-18T09:30:00Z",
    "operating_state_ref": "current-capacity"
  },
  "output": {
    "controls_decision_ref": "controls:lqp-20260518-093243",
    "compliance_decision_ref": "compliance-decision:req-20260518-093241",
    "lqp_id": "lqp-20260518-093243",
    "decision": "approved",
    "checks": {
      "data_scale": {
        "estimate": 412000,
        "limit": 50000000,
        "result": "pass"
      },
      "complexity": {
        "score": 4,
        "limit": 50,
        "result": "pass"
      },
      "classification": {
        "required": "INTERNAL",
        "ceiling": "INTERNAL",
        "result": "pass"
      },
      "compliance": {
        "metric_signal": false,
        "purpose_signal": false,
        "result": "standard"
      },
      "concurrency": {
        "active": 3,
        "limit": 20,
        "result": "pass"
      }
    },
    "timeout_seconds": 30
  }
}

```

An approved controls decision releases the Logical Query Plan and execution budget to the PQP.

Physical Query Planner (PQP)

Governing principles: P1 - Semantic abstraction and P10 - Deterministic computation, not generation

Business definition. The PQP translates an approved business calculation into bounded instructions that registered data sources can execute.

Why it exists. Business definitions should remain stable when physical schemas or source technologies change. The PQP contains that translation boundary and keeps physical details away from AI consumers and language models.

Input and output. It receives the approved Logical Query Plan, execution budget, pinned mappings, and source capabilities. It outputs a physical execution envelope containing source sub-plans, protection directives, and integrity evidence.

The PQP is the boundary between logical concepts and physical execution. It resolves the approved physical mapping associated with each pinned definition version, expands logical time expressions, groups work by source affinity, and compiles source-specific sub-plans.

The PQP applies row predicates and masking requirements to each relevant sub-plan. Where a source can enforce masking, the generated query performs it before values leave that source. Otherwise, an approved controlled execution stage applies the mask before returning data outside the FQE boundary.

A repeatable physical plan requires the same LQP, definition and mapping versions, planner version, configuration, and source capabilities. The PQP records those dependencies. It does not hold source credentials or execute plans.

Compilation Stages

Stage	Responsibility
Mapping resolution	Resolve each logical scan and measure against the mapping pinned to the approved definition version.
Predicate binding	Bind typed dimension filters and RAPL row scopes using backend-safe parameters.
Time resolution	Expand relative periods against the recorded evaluation timestamp and calendar.
Source grouping	Group compatible work by data affinity and identify required cross-source assembly.
Source instruction compilation	Produce a bounded sub-plan in the registered instruction format for each source group.
Plan sealing	Create an integrity digest and record planner, instruction-format, mapping, and configuration versions.

An implementation may delegate federation to one execution service or create several source sub-plans with an explicit assembly contract. Both approaches implement the same logical responsibility; Appendix B proposes a specific choice.

Input and Output Example

The PQP resolves the approved mappings and creates a technology-neutral execution envelope:

```
{
  "input": {
    "lqp_id": "lqp-20260518-093243",
    "controls_decision_ref": "controls:lqp-20260518-093243",
    "mapping_set_ref": "approved-mappings",
    "source_capabilities_ref": "registered-source-capabilities"
  },
  "output": {
    "plan_id": "plan-20260518-093244",
    "lqp_id": "lqp-20260518-093243",
    "controls_decision_ref": "controls:lqp-20260518-093243",
    "entitlement_projection_ref": "projection:req-20260518-093241",
    "mapping_versions": [
      "portfolio-mapping-4.6"
    ],
    "plan_digest": "digest:<illustrative-plan-value>",
    "sub_plans": [
      {
        "source_id": "portfolio-performance-source",
        "metrics": [
          "portfolio_return",
          "benchmark_return"
        ],
        "filters": [
          "asset_class",
          "authorized_portfolios",
          "resolved_date_range"
        ]
      }
    ],
    "resolved_date_range": {
      "from": "2026-04-01",
      "to": "2026-05-18"
    },
    "column_masks": [],
    "timeout_seconds": 30
  }
}
```

The physical execution envelope becomes the FQE input; no upstream component executes against a data source.

Federated Query Engine (FQE)

Governing principles: P1 - Semantic abstraction, P4 - Complete analytical lineage, and P10 - Deterministic computation, not generation

Business definition. The FQE is the controlled execution boundary that runs approved source instructions and assembles their results.

Why it exists. Source credentials, timeouts, federation, caching, masking, and failure handling require one accountable boundary. Concentrating them here prevents upstream components and AI consumers from gaining direct backend access.

Input and output. It receives an approved physical execution envelope. It outputs a typed result or an explicit terminal outcome, plus execution evidence for the ALS.

The FQE is the only logical component with access to registered backend connections. It validates source availability, executes approved sub-plans within the assigned timeout, joins compatible results on governed dimensions, enforces any remaining masks within its controlled boundary, and records execution outcomes.

Caching uses a canonical key that includes the effective plan, tenant, entitlement projection, definition versions, and relevant configuration. A cache hit still passes current authorization and produces a new lineage event. Compliance-purpose queries bypass the cache unless approved policy provides equivalent evidence and freshness guarantees.

Source failure is visible. The response identifies incomplete sources and affected metrics. A partial result may be returned only when the approved operation contract permits partial semantics; otherwise, the request fails rather than representing missing values as a complete answer.

The FQE records source identifiers, plan hashes, timing, row counts, cache status, errors, and mask application. It does not expose credentials or unrestricted physical schema details to consumers or language models.

Execution Pipeline

Stage	Required Behavior
Plan admission	Verify plan integrity, source registration, timeout, tenant, and control approval.
Cache evaluation	Check an entitlement-safe canonical key and current policy before reuse.
Source execution	Run sub-plans within their time and resource budgets.
Assembly	Join or combine results only on approved compatible dimensions.
Final protection	Apply any remaining masks and verify the returned schema before release.

Stage	Required Behavior
Evidence write	Record the outcome before returning the result or terminal failure.

Supported source categories may include relational analytical stores, governed semantic services, approved data APIs, relationship stores, and multidimensional analytical engines. Technical connectivity does not make a source usable automatically. Each connection requires approved mappings, credentials, classifications, timeout behavior, and failure semantics.

Result-size limits determine whether the engine returns inline data, paginates an approved dataset, streams within a controlled protocol, or rejects the request. These choices are part of the operation contract and response evidence.

Input and Output Example

The FQE executes the approved plan and returns typed result rows:

```

{
  "input": {
    "plan_id": "plan-20260518-093244",
    "controls_decision_ref": "controls:lqp-20260518-093243"
  },
  "output": {
    "result_id": "res-20260518-093247",
    "data_ref": "result:res-20260518-093247",
    "request_id": "req-20260518-093241",
    "plan_id": "plan-20260518-093244",
    "status": "complete",
    "execution_profile": "full_analytical",
    "sources_used": [
      "portfolio-performance-source"
    ],
    "cache_status": "miss",
    "schema": [
      {
        "field": "portfolio",
        "type": "string",
        "label": "Portfolio"
      },
      {
        "field": "portfolio_return",
        "type": "decimal",
        "label": "Portfolio Return",
        "unit": "ratio"
      },
      {
        "field": "benchmark_return",
        "type": "decimal",
        "label": "Benchmark Return",
        "unit": "ratio"
      }
    ],
    "rows": [
      {
        "portfolio_id": "GLOB_EQ_OPP",
        "portfolio": "Global Equity Opportunities",
        "portfolio_return": 0.0421,
        "benchmark_return": 0.0385
      },
      {
        "portfolio_id": "ASIA_PAC_GRW",
        "portfolio": "Asia Pacific Growth",
        "portfolio_return": 0.0367,
        "benchmark_return": 0.039
      },
      {
        "portfolio_id": "UK_CORE_INC",
        "portfolio": "UK Core Income",
        "portfolio_return": 0.0287,
        "benchmark_return": 0.0254
      },
      {
        "portfolio_id": "EUR_BAL_INC",
        "portfolio": "EUR Balanced Income",
        "portfolio_return": 0.0193,
        "benchmark_return": 0.0231
      }
    ]
  }
}

```

```
]
}
}
```

The typed result now branches to presentation, response assembly, and evidence recording. DVL determines how consumers should display it.

Data Visualization Language (DVL)

Governing principle: P7 - Deterministic visualization

Business definition. The DVL turns a governed result into a consistent chart or table contract that consumers can render.

Why it exists. Presentation choices can change interpretation. A governed display contract prevents each consumer or language model from independently choosing labels, units, ordering, thresholds, and chart form.

Input and output. It receives the approved intent, typed result schema, values, and presentation metadata. It outputs a versioned display specification or the governed table fallback.

The DVL produces a structured chart or table specification from the approved intent and result schema. A registry maps comparison, trend, distribution, threshold, attribution, relationship, and composition patterns to compatible presentation contracts. A governed table is the fallback.

Contract selection is deterministic for the same intent, result schema, registry version, and configuration. Labels and units come from approved definitions, not physical fields. An authorized analyst may request an allowed compatible override; the platform rejects incompatible choices and records accepted overrides.

The DVL returns a specification, not a rendered image. Consumer applications render it directly or pass it to an independently governed rendering service.

Intent Pattern Taxonomy

Pattern	Purpose	Typical Contract
Comparison	Compare measures across discrete categories.	Grouped bar chart or governed table.
Trend	Show change over an ordered time dimension.	Line chart.
Distribution	Show spread, frequency, or concentration.	Histogram or governed table.
Threshold	Compare observations with a limit or benchmark.	Threshold matrix, bar chart, or table.
Attribution	Decompose a total into contributing factors.	Waterfall chart.
Relationship	Show association between two measures.	Scatter plot.
Composition	Show part-to-whole structure.	Treemap or governed table.

Chart Contract Selection

Each registered contract declares compatible patterns, required field types, cardinality limits, axis or column bindings, sorting, interaction behavior, accessibility requirements, and fallback behavior. The selector evaluates the most specific compatible contract first. If none matches, it returns the governed table contract rather than asking a language model to invent a display.

Themes may change approved colors and typography without changing the analytical contract. Threshold colors, units, labels, ordering, and benchmark semantics come from governed metadata and must remain interpretable in accessible and monochrome presentations.

Input and Output Example

The DVL selects the registered comparison contract from the intent and result shape:

```
{
  "input": {
    "result_id": "res-20260518-093247",
    "data_ref": "result:res-20260518-093247",
    "intent_pattern": "comparison",
    "presentation_metadata_ref": "smr:presentation:portfolio-benchmark-comparison@2.0"
  },
  "output": {
    "display_spec_ref": "display:res-20260518-093247",
    "result_id": "res-20260518-093247",
    "type": "chart",
    "contract": "multi_series_comparison",
    "contract_version": "2.0",
    "mark": "bar",
    "category": {
      "field": "portfolio",
      "label": "Portfolio"
    },
    "measures": [
      {
        "field": "portfolio_return",
        "label": "Portfolio Return",
        "format": "percentage"
      },
      {
        "field": "benchmark_return",
        "label": "Benchmark Return",
        "format": "percentage"
      }
    ],
    "sort": {
      "field": "portfolio_return",
      "direction": "descending"
    }
  }
}
```

In parallel with DVL, the NSA receives the bounded result content needed for an optional narrative.

Narrative Synthesis Agent (NSA)

Governing principles: P6 - Governed narrative and P10 - Deterministic computation, not generation

Business definition. The NSA provides an optional plain-language explanation of a completed result.

Why it exists. A concise explanation makes analysis easier to consume, but narrative generation must remain separate from calculation and must not introduce unsupported values or conclusions.

Input and output. It receives only approved result labels, values, units, dimensions, and permitted derivations. It outputs a validated narrative or an explicit omission state.

The NSA optionally produces a concise plain-language summary after computation. Its input is limited to approved result labels, values, units, and dimensions. It does not calculate metrics or change the result.

A validator checks every stated number against the returned data or an independently recomputed permitted derivation. If validation fails, the service may retry once and then omits the narrative while recording the failure. Quantitative matching reduces unsupported claims but does not prove that wording, emphasis, or causal interpretation is correct. Evaluation therefore includes semantic review, not numeric matching alone.

Disabling the NSA has no effect on calculation, display specification, lineage, or compliance evidence.

Anchoring and Validation

The narrative contract separates a short lead statement, supporting detail, and the result identifiers to which each claim is anchored. Validation covers numbers, units, dimension labels, comparison direction, and permitted derived counts. It rejects invented causes, recommendations, forecasts, or significance claims unless the approved operation explicitly provides those outputs.

The lineage record captures the model, prompt-template version, validation outcome, retry count, and omission reason. Model selection may vary by result complexity, but changing models must not alter computed data or the DVL contract.

Input and Output Example

The NSA summarizes only statements supported by the result rows:

```

{
  "input": {
    "result_id": "res-20260518-093247",
    "data_ref": "result:res-20260518-093247",
    "allowed_content": [
      "approved labels",
      "result values",
      "units",
      "permitted derived counts"
    ]
  },
  "output": {
    "narrative_ref": "narrative:res-20260518-093247",
    "result_id": "res-20260518-093247",
    "narrative": {
      "lead": "Two of four equity portfolios outperformed their benchmarks this quarter.",
      "detail": "Global Equity Opportunities returned 4.21% compared with 3.85% for its benchmark. UK Core Income returned 2.87% compared with 2.54%.",
      "anchored_to": [
        "GLOB_EQ_OPP",
        "UK_CORE_INC",
        "ASIA_PAC_GRW",
        "EUR_BAL_INC"
      ]
    },
    "validation": {
      "numbers": "passed",
      "units": "passed",
      "dimension_labels": "passed"
    }
  }
}

```

DVL and NSA write their presentation evidence to the ALS, where it joins the request, decision, plan, and execution events already recorded for the request.

Analytical Lineage Store (ALS)

Governing principles: P4 - Complete analytical lineage and P8 - Explainability at every layer

Business definition. The ALS is the evidence ledger for how each governed request was interpreted, authorized, controlled, planned, executed, and presented.

Why it exists. A result cannot be traced or assessed for repeatability if its definitions, permissions, decisions, source references, and terminal outcome are scattered or overwritten.

Input and output. It receives correlated events from each stage. It outputs an authorized lineage chain, searchable references, integrity evidence, and the event set used by PAS and audit export.

The ALS preserves correlated events for every accepted request, including entitlement decisions, controls decisions, execution, presentation, and terminal failure. Records use a common request identifier so an authorized reviewer can reconstruct the sequence without treating one final payload as the complete history.

By default, lineage stores hashes, definition and policy versions, bounded summaries, source references, plan identifiers, decisions, and timing rather than full raw requests or results. An approved policy may require additional content. The design minimizes sensitive data while preserving evidence for its intended review.

Claims of immutability depend on storage enforcement. Write-once retention, versioning, signatures, encryption, tenant-scoped authorization, key management, reconciliation, and tested recovery provide the required integrity properties. Application convention alone is insufficient. Corrections create linked amendment records rather than altering prior evidence.

Retention varies by record class, jurisdiction, and business purpose. Definition versions and references needed to interpret retained results remain available for at least the corresponding evidence period.

Stored Event Types

Event	Minimum Evidence
Request accepted	Tenant, caller reference, tool, operation, parameters or bounded request representation, and timestamp.
Intent resolved	Candidate set, selected operation, model evidence, confidence, and confirmation history.
Entitlement projected	Policy versions, active roles, resolved row scopes, masks, ceiling, and decision.
Semantic request validated	Pinned definitions, entitlement projection, compliance inputs, and logical plan digest.
Controls evaluated	Each control input, limit, outcome, reason, and assigned timeout.

Event	Minimum Evidence
Plan compiled	Logical and physical plan digests plus definition, mapping, planner, and instruction-format versions.
Execution completed	Sources, timings, row counts, cache status, failures, and result digest.
Presentation prepared	DVL contract, narrative validation, and any omission reason.
Provenance artifact sealed	Artifact identifier, schema version, verification metadata, and export state when PAS is required.
Response assembled	Response status, included references, warnings, errors, and delivery outcome.
Request terminated	Rejection, cancellation, timeout, or failure stage and structured reason.

Isolation, Retention, and Audit Export

Every index entry and object key is tenant-scoped. Authorization applies to both search metadata and stored objects. Audit export is a separately authorized capability; possession of a result identifier does not grant access to its evidence.

Retention policy defines periods for request events, controls decisions, execution evidence, result artifacts, blocked requests, and definition versions. A shorter result-data period may be appropriate, but retained evidence must state when source data or result payloads are no longer available for exact reconstruction. Legal hold, deletion, amendment, and key-rotation behavior require explicit design and testing.

An audit export includes the selected event chain, referenced definition and policy versions, verification material, export scope, and exporter's identity. The platform signs the export package and records the export as a new auditable event.

Input and Output Example

The ALS correlates stage records without storing the full result by default:

```

{
  "input": {
    "request_id": "req-20260518-093241",
    "resolved_request_id": "resolved-20260518-093242",
    "entitlement_projection_ref": "projection:req-20260518-093241",
    "lqp_id": "lqp-20260518-093243",
    "controls_decision_ref": "controls:lqp-20260518-093243",
    "compliance_decision_ref": "compliance-decision:req-20260518-093241",
    "plan_id": "plan-20260518-093244",
    "result_id": "res-20260518-093247",
    "display_spec_ref": "display:res-20260518-093247",
    "narrative_ref": "narrative:res-20260518-093247"
  },
  "output": {
    "lineage_ref": "lineage:req-20260518-093241",
    "request_id": "req-20260518-093241",
    "result_id": "res-20260518-093247",
    "events": [
      {
        "type": "request_accepted",
        "record_id": "evt-request-001"
      },
      {
        "type": "intent_resolved",
        "record_id": "evt-intent-001"
      },
      {
        "type": "entitlement_projected",
        "record_id": "evt-access-001"
      },
      {
        "type": "semantic_validated",
        "record_id": "evt-validation-001"
      },
      {
        "type": "controls_evaluated",
        "record_id": "evt-controls-001"
      },
      {
        "type": "plan_compiled",
        "record_id": "evt-plan-001"
      },
      {
        "type": "execution_completed",
        "record_id": "evt-execution-001"
      },
      {
        "type": "presentation_prepared",
        "record_id": "evt-presentation-001"
      }
    ],
    "result_digest": "digest:<illustrative-value>",
    "retention_policy": "analytical-evidence-standard"
  }
}

```

When both compliance signals are active, the correlated ALS event set becomes the PAS input.

Provenance Artifact Service (PAS)

Governing principles: P2 - Controls before execution, P4 - Complete analytical lineage, and P9 - Administrator sovereignty within governance bounds

Business definition. PAS packages the evidence required for a compliance-purpose result into a sealed, independently verifiable artifact.

Why it exists. Standard lineage supports operational review, while regulated uses may require a fixed schema, named framework references, a digital signature, and an export gate.

Input and output. It receives the active compliance trigger and the correlated ALS event set. It outputs an `artifact_state_ref` for a sealed artifact with verification metadata and an explicit export state, or a failure state that keeps export blocked.

PAS assembles additional evidence for a request whose two compliance signals are active. It reads the relevant ALS events, adds the metric, framework, intent, policy, plan, execution, and result references required by the approved artifact schema, and signs the canonical artifact bytes.

The platform enforces the export gate. A compliance-purpose result cannot be exported until PAS confirms that the artifact is complete and sealed. The response identifies the artifact, schema version, signing key identifier, algorithm, and verification status.

A valid signature shows that the signed bytes have not changed since sealing. It does not prove that the calculation is correct or that the artifact satisfies a legal requirement. The organization validates the schema, retention, signing controls, and approval process for each intended framework.

Compliance Artifact Contract

Field	Purpose
compliance purpose and intent score	Records the purpose input and threshold decision.
triggered metrics and frameworks	Identifies the approved metadata that activated the additional evidence path.
regulatory trace identifier	Correlates the artifact with the retained event chain.
artifact schema version	Identifies the structure used by reviewers and verification software.
definition, policy, plan, and result references	Binds the artifact to the governed computation.
signing key identifier, algorithm, and signature	Supports independent integrity verification without exposing signing-key material.
export status	Shows whether sealing completed and export is permitted.

If assembly or signing fails, the result remains non-exportable and the ALS records the terminal artifact state. Amendments create a new signed artifact that references the original; they do not replace sealed evidence.

Running Example: No PAS Invocation

The portfolio comparison reaches the compliance decision with both signals inactive. PAS is not invoked. The orchestration path records that routing outcome, and response assembly uses the SCL decision to produce the explicit standard compliance state:

```
{
  "input": {
    "request_id": "req-20260518-093241",
    "compliance_decision_ref": "compliance-decision:req-20260518-093241",
    "compliance_trigger": {
      "metric_signal": false,
      "purpose_signal": false
    }
  },
  "output": {
    "request_id": "req-20260518-093241",
    "pas_invoked": false,
    "artifact_required": false,
    "artifact_id": null,
    "compliance_route": "standard",
    "reason": "two_signal_trigger_not_active"
  }
}
```

The result, presentation outputs, lineage reference, compliance decision, and any sealed artifact state now become the response-assembly input.

Structured Response

Business definition. The structured response is the governed delivery contract between the analytical platform and its consumers.

Why it exists. Consumers need one unambiguous way to distinguish data, presentation, narrative, evidence, warnings, compliance state, and terminal status without reconstructing analytical meaning.

Input and output. Response assembly receives the typed result or terminal outcome plus presentation and evidence references. It outputs one structured envelope that the consumer can render, retain, or pass to an authorized follow-up action.

The platform is headless. It returns a structured response that lets different consumers present the same governed result without reinterpreting its meaning.

Response Element	Purpose
request_id and result_id	Correlate the response, follow-up actions, and evidence.
status	Distinguishes complete, incomplete, rejected, failed, and timed-out outcomes.
data	Contains typed, paginated rows or an approved dataset extract.
display_spec	Contains the DVL chart or table contract when required.
narrative	Contains the optional validated NSA summary.
lineage	References the authorized analytical evidence.
compliance	Reports artifact and export-gate state when applicable.
warnings and errors	Identify limitations, affected sources, and remediation.

DVL Specification Format

A DVL specification is a discriminated chart or table envelope. It contains approved labels, units, formatting hints, encodings or columns, interaction rules, and the contract version. It may contain result data or a controlled reference, depending on response size and consumer capability. It never substitutes physical field names for governed labels.

Response Status and Failure Semantics

Complete, incomplete, rejected, failed, and timed-out are distinct states. An incomplete response identifies missing sources or metrics and appears only when the operation contract permits partial semantics. A rejection means a governance or validation rule prevented execution. A failure or timeout means approved execution did not complete. Consumers must present these states explicitly and must not render an incomplete or failed response as a complete governed answer.

Input and Output Example

The final response combines the computed result with its governed presentation and evidence references:

```

{
  "input": {
    "request_id": "req-20260518-093241",
    "result_id": "res-20260518-093247",
    "data_ref": "result:res-20260518-093247",
    "display_spec_ref": "display:res-20260518-093247",
    "narrative_ref": "narrative:res-20260518-093247",
    "lineage_ref": "lineage:req-20260518-093241",
    "compliance_decision_ref": "compliance-decision:req-20260518-093241",
    "artifact_state_ref": null
  },
  "output": {
    "request_id": "req-20260518-093241",
    "result_id": "res-20260518-093247",
    "status": "complete",
    "data_ref": "result:res-20260518-093247",
    "display_spec_ref": "display:res-20260518-093247",
    "narrative_ref": "narrative:res-20260518-093247",
    "lineage": {
      "record_ref": "lineage:req-20260518-093241",
      "available_to_caller": true
    },
    "compliance": {
      "state_ref": "compliance:req-20260518-093241",
      "triggered": false,
      "export_status": "permitted"
    },
    "warnings": [],
    "errors": []
  }
}

```

The MCP Capability Layer returns this response to the consumer, completing the governed request flow.

Governed Data-Mining Example

The portfolio-comparison example demonstrates governed metric calculation. Data mining uses the same platform controls but resolves an approved dataset contract rather than a metric formula. The output is a bounded, typed extract whose selection rules, permitted fields, source state, membership, pagination, and export decision are recorded.

In this variation, an automated research application already knows the approved operation and submits a structured request. It bypasses IRA language interpretation, but it does not bypass SMR resolution, RAPL, SVL, SCL, PQP, FQE, or ALS. The approved operation uses the `data_retrieval` execution profile, so DVL and NSA are not invoked. PAS remains independent of the profile and would run only if the compliance trigger were active.

Component Handoffs

Component	Data-Mining Responsibility	Contract Handoff
AI consumer	Requests a known governed extract for downstream research.	<code>operation_id</code> , typed parameters, requested fields, and authenticated context to MCP.
MCP and SMR	Resolve the approved operation and dataset contract.	Pinned operation and dataset-definition references to RAPL and SVL.
RAPL and DES	Limit the extract to permitted datasets, fields, portfolios, classifications, and export rights.	Entitlement projection containing row scope, field projection, masks, and export policy.
SVL	Validate the dataset contract, parameters, date range, field compatibility, and stable ordering.	Backend-independent LQP describing the governed dataset selection.
SCL	Evaluate estimated row volume, complexity, classification, compliance purpose, concurrency, and timeout.	Approved execution budget and compliance decision, or structured rejection.
PQP and FQE	Bind approved mappings, execute the bounded selection, apply protections, and paginate the result.	Typed dataset pages plus the extract manifest, source snapshot reference, and execution evidence.
ALS	Correlate the contract, projection, plan, snapshot, page set, membership digest, export decision, and terminal outcome.	Authorized lineage reference for review and repeatability testing.
Structured Response	Return the extract without adding a chart or AI narrative.	Data reference, schema, pagination, extract manifest, lineage, and export status.

Technology-Neutral Contract Example

The application requests daily fixed-income positions for its authorized portfolios during April 2026. The requested fields are a subset of the approved dataset contract:

```

{
  "input": {
    "request_id": "req-20260518-101500",
    "operation_id": "retrieve_position_history",
    "operation_definition_ref": "smr:operation:retrieve_position_history@1.0.0",
    "dataset_definition_ref": "smr:dataset:fixed_income_daily_positions@1.0.0",
    "execution_profile": "data_retrieval",
    "parameters": {
      "portfolio_scope": "caller_authorized_portfolios",
      "asset_class": "FIXED_INCOME",
      "date_from": "2026-04-01",
      "date_to": "2026-04-30",
      "requested_fields": [
        "portfolio_id",
        "instrument_id",
        "position_date",
        "market_value",
        "duration",
        "currency"
      ],
      "page_size": 10000
    },
    "entitlement_projection": {
      "entitlement_projection_ref": "projection:req-20260518-101500",
      "authorized_portfolios": [
        "GLOB_FI_CORE",
        "US_CREDIT_OPP"
      ],
      "approved_fields": [
        "portfolio_id",
        "instrument_id",
        "position_date",
        "market_value",
        "duration",
        "currency"
      ],
      "classification_ceiling": "INTERNAL",
      "export_policy_ref": "export-policy:research-internal@3.1"
    }
  },
  "output": {
    "request_id": "req-20260518-101500",
    "result_id": "dataset-res-20260518-101512",
    "status": "complete",
    "data_ref": "dataset:dataset-res-20260518-101512",
    "dataset_definition_ref": "smr:dataset:fixed_income_daily_positions@1.0.0",
    "source_snapshot_ref": "snapshot:portfolio-source:2026-05-01T00:00:00Z",
    "selection_digest": "digest:<illustrative-selection-value>",
    "dataset_membership_digest": "digest:<illustrative-membership-value>",
    "schema": [
      {
        "field": "portfolio_id",
        "type": "string"
      },
      {
        "field": "instrument_id",
        "type": "string"
      },
      {
        "field": "position_date",

```

```

    "type": "date"
  },
  {
    "field": "market_value",
    "type": "decimal",
    "unit": "reporting_currency"
  },
  {
    "field": "duration",
    "type": "decimal",
    "unit": "years"
  },
  {
    "field": "currency",
    "type": "string"
  }
],
"ordering": [
  "position_date",
  "portfolio_id",
  "instrument_id"
],
"pagination": {
  "page_size": 10000,
  "total_rows": 24837,
  "next_page_token": "page:dataset-res-20260518-101512:2"
},
"extract_manifest_ref": "manifest:dataset-res-20260518-101512",
"lineage_ref": "lineage:req-20260518-101500",
"export": {
  "status": "permitted",
  "policy_ref": "export-policy:research-internal@3.1"
},
"warnings": [],
"errors": []
}
}

```

The extract manifest covers every page and records the dataset contract, logical selection, effective field and row projection, stable ordering, source snapshot, per-page digests, complete membership digest, and export decision. An authorized rerun can therefore distinguish a changed selection rule from changed dataset membership.

The platform's accuracy claim for this workflow is limited to the governed extract: that the returned records and fields conform to the approved contract, permissions, source state, and selection rules. Feature engineering, statistical analysis, model training, model validation, and conclusions produced from the extract remain separate governed activities with their own evidence and approval requirements.

External Components

External components remain outside the AI Analytics Platform boundary and retain their own ownership and controls.

Conversational AI - Chat Front End

Business definition and purpose. The chat front end is a consumer experience for people who request analysis conversationally. It exists to manage dialogue and presentation without owning analytical meaning or controls.

Input and output. It receives a user's question, confirmation choices, and structured platform responses. It outputs authenticated analytical requests, rendered answers, and governed follow-up actions such as drilldown. It does not resolve definitions, enforce entitlements, plan queries, or gain access to backend credentials and schemas.

Semantic Data Repository (SDR)

Business definition and purpose. The SDR is the governed description of the organization's data assets. It exists so analytical definitions can connect to understood data structures, quality expectations, and lineage rather than undocumented source fields.

Input and output. It receives approved data models, critical data elements, quality rules, structural metadata, and data-lineage references. It outputs versioned context and mapping targets used by SMR definitions and physical planning. The proposal does not assume that every organization already has a suitable SDR; readiness assessment establishes its coverage, ownership, freshness, and mapping quality.

Data Entitlements Store (DES)

Business definition and purpose. The DES is the independent policy source for who may use which analytical concepts and data populations. It exists to keep access decisions separate from metric authorship and platform administration.

Input and output. It receives approved policies for metrics, dimensions, row scopes, masks, and classification ceilings. It outputs versioned role and policy definitions to RAPL. Policies reference logical concepts rather than physical table and column names.

External Language Model Service

Business definition and purpose. The external language model service provides bounded language interpretation and synthesis. It exists to support natural interaction while remaining outside calculation, authorization, controls, and source execution.

Input and output. For IRA, it receives approved candidate descriptions and the user's question and returns ranking and parameter suggestions. For NSA, it receives bounded result content and returns a draft narrative. IRA and NSA validate those outputs before the flow continues.

Registered Data Sources

Business definition and purpose. Registered data sources hold the governed business data on which approved analytics operate. They remain systems of record or analytical services rather than becoming part of the AI layer.

Input and output. They receive bounded, authorized source instructions from the FQE and return typed data or explicit failures. They do not receive natural-language questions, model prompts, or consumer credentials.

Optional Rendering Service

Business definition and purpose. The optional rendering service converts a governed display contract into a static visual for consumers that cannot render it directly. It exists as a presentation utility, not as part of analytical computation.

Input and output. It receives a self-contained display specification and outputs SVG or PNG. It has no access to analytical definitions, source credentials, or execution backends. Appendix B identifies an illustrative product and implementation for this role.

Design Decisions Requiring Evaluation

The proposal leaves several choices for implementation and governance teams to validate:

- role combinations and separation of duties;
- interaction between explicit denies and multi-role grants;
- operation contracts that may return partial results;
- evidence and freshness rules for caching;
- source snapshot retention and reconstruction;
- supported formula operations and source instruction formats;
- confidence and compliance-purpose thresholds;
- storage controls for evidence integrity and retention; and
- artifact schemas and approval processes for each intended regulatory use.

These are evaluation questions, not hidden implementation details. The evaluation plan in Appendix B defines how to test the architecture before broader adoption.

Governed AI-Enabled Analytics and Data Mining - Technical White Paper - Draft

3. Glossary

This glossary defines the principal components, contracts, roles, and technical terms used in the AI Analytics Platform white paper. Logical architecture terms follow Section 2. Product-specific terms are identified as elements of the illustrative reference implementation in Appendix B. Terms are sorted alphabetically, and cross-references to other entries are shown in **bold**.

Term	Abbreviation	Definition
Admin API	N/A	An illustrative Appendix B service for governed administration of platform settings and authoring workflows. It is not an analytical execution path and does not grant analytical data access. Data Entitlements Store (DES) policies remain under the Entitlements Manager's control.
AI Analytics Platform	N/A	The proposed architecture for governed AI-enabled analytics and data mining. It combines bounded language interpretation with approved semantic definitions, deterministic planning and computation, entitlement enforcement, controls, structured presentation, and retained evidence.
AI consumer	N/A	A conversational assistant, autonomous agent, data-mining workflow, or custom application that submits an authenticated analytical request and presents the governed response. A consumer does not own analytical definitions, permissions, query planning, or governed computation.
Analytical Lineage Store	ALS	The evidence ledger for accepted analytical requests. It correlates request, intent, entitlement, validation, controls, planning, execution, presentation, artifact, response, and terminal events. Its integrity depends on enforced retention, authorization, signing or content addressing, reconciliation, and recovery controls rather than application convention alone.

Term	Abbreviation	Definition
Authentication and identity token	N/A	A host-issued credential carried through trusted request transport and validated at platform ingress. Verified identity claims are made available through server-side request context; the bearer token is not an analytical tool argument and is not exposed to a language model. Appendix B illustrates this contract with a JSON Web Token (JWT) .
Compliance trigger	N/A	The deterministic decision that additional provenance evidence is required. The proposed rule requires both a compliance-relevant metric signal and a compliance-purpose signal. A natural-language request receives the purpose signal from the Intent Resolution Agent (IRA) subject to threshold evaluation; a structured caller declares purpose through a validated field. Either signal alone is insufficient.
Data Context Store	DCS	The shared versioned storage and access service that houses the separate Semantic Metrics Repository (SMR) and Semantic Data Repository (SDR) in the illustrative architecture. It does not contain every platform policy or runtime record; the DES and ALS remain distinct responsibilities.
Data Entitlements Store	DES	The independent policy source for analytical access. It stores versioned policy for permitted metrics, dimensions, row scopes, column masks, and classification ceilings. The MCP layer uses a bounded policy view for catalog discovery, while the Role-Aware Projection Layer (RAPL) performs the full request-time entitlement evaluation.

Term	Abbreviation	Definition
Data mining	N/A	In this paper, the use of governed bulk datasets by an authorized research application, agent, or analytical process to investigate patterns or develop models. The platform controls dataset selection, fields, row scope, source state, pagination, export, and evidence. Downstream feature engineering, statistical analysis, model training, validation, and conclusions remain separately governed activities.
Data Visualization Language	DVL	The governed presentation contract produced deterministically from the approved intent, result schema, and registered presentation metadata. It returns a chart or table specification rather than a rendered image. An authorized caller may request a compatible governed override; incompatible presentation choices are rejected.
Dataset contract	N/A	An approved, versioned <code>analytical_dataset</code> definition in the SMR . It declares the permitted fields, classifications, selection rules, source mapping, ordering, pagination, and freshness expectations for governed data retrieval.
Execution profile	N/A	A field on an approved <code>analytical_operation</code> definition that controls optional response assembly after the mandatory governed pipeline has run. <code>data_retrieval</code> returns a governed dataset, <code>metric_query</code> adds a DVL specification, and <code>full_analytical</code> may also add an NSA narrative. The profile never bypasses entitlements, validation, controls, planning, execution, or lineage. PAS invocation is controlled by the compliance trigger , not by the execution profile.

Term	Abbreviation	Definition
External language model service	N/A	A model service called through bounded adapters by the IRA or NSA . It may rank approved candidates, suggest parameter bindings, or draft a narrative from permitted result content. It does not receive source credentials, authorize access, calculate governed metrics, or generate the executable physical query.
Federated Query Engine	FQE	The controlled execution boundary and the only logical component with registered backend connections and credentials. It admits approved physical execution envelopes, runs bounded source instructions, assembles compatible results, applies any remaining protections, and records execution evidence.
Financial Services Reference Model	FSRM	Illustrative Appendix B bundles of proposed SMR definitions for shared dimensions and selected financial-services domains. The bundles are seeded in proposed state and become resolvable only after governance approval.
Intent confirmation card	N/A	A structured choice returned when intent or compliance purpose is ambiguous. It presents ranked approved operations, bound parameters, material filters, and a presentation preview without executing a backend query or exposing physical schemas.
Intent Resolution Agent	IRA	The bounded AI component that translates natural language into an approved operation and typed parameters. It ranks candidates from an entitlement-scoped SMR catalog view, records confidence and purpose evidence, and requests confirmation when meaning is uncertain. It does not receive credentials, generate executable queries, or approve definitions. Structured requests that already identify an approved operation bypass the IRA.

Term	Abbreviation	Definition
JSON Web Token	JWT	The token format used for authentication and identity in the illustrative Appendix B implementation. The implementation validates its signature, issuer, audience, expiry, and required claims against trusted keys before analytical processing begins.
Knowledge Store	N/A	An illustrative Appendix B store of versioned, non-user-data guidance exposed read-only through MCP resources. Knowledge resources help consumers use the platform but do not provide an alternate analytical execution path.
Logical Query Plan	LQP	A backend-independent plan produced by the Semantic Validation Layer (SVL) . It carries pinned definition versions, logical calculations, filters, row scope, masks, compliance inputs, output shape, and other governed planning context. It contains no credentials, connection endpoints, physical table names, or executable SQL.
MCP Capability Layer	MCP	The single governed platform entry point. It authenticates requests, applies entitlement-aware tool and catalog discovery, accepts natural-language or structured operation requests, correlates the request, and returns the Structured Response . A structured request may bypass the IRA, but it still resolves an approved SMR operation and passes the complete entitlement and controls pipeline.
Metric definition	N/A	An approved, versioned SMR declaration of a metric's business meaning, formula or governed measure, aggregation behavior, units, compatible dimensions, classification, source affinity, mapping reference, owner, and tests. Access policy is maintained independently in the DES .

Term	Abbreviation	Definition
Model Context Protocol	MCP	The protocol used by the proposed platform to expose governed tools, resources, and prompts to AI consumers. The protocol does not itself provide semantic governance, entitlement enforcement, or analytical correctness.
Narrative	N/A	An optional plain-language explanation produced by the Narrative Synthesis Agent (NSA) from approved result labels, values, units, dimensions, and permitted derivations. Unsupported numbers, causes, recommendations, forecasts, or significance claims are rejected or omitted.
Narrative Synthesis Agent	NSA	The optional post-computation AI component that drafts a result-grounded narrative. A deterministic validator checks numbers, units, labels, comparison direction, and permitted derivations before release. Failure to produce a valid narrative does not change the computed data, DVL specification, lineage, or compliance evidence.
Optional Rendering Service	N/A	An external presentation utility that converts a self-contained DVL specification into SVG or PNG. It has no access to analytical definitions, source credentials, or execution backends and does not participate in governed computation.
Physical Query Planner	PQP	The deterministic boundary that resolves approved mappings and source capabilities for an LQP and produces a bounded physical execution envelope. It binds typed predicates, expands time periods, groups work by source, applies protection directives, and records plan integrity evidence without holding source credentials or executing the plan.

Term	Abbreviation	Definition
Provenance Artifact	N/A	A sealed, cryptographically signed evidence package produced when the compliance trigger is active. It binds the request and purpose evidence to the applicable definitions, entitlement decision, controls, plans, execution, result, and verification metadata. A valid signature establishes that the sealed bytes have not changed; it does not prove that the calculation is correct or legally sufficient.
Provenance Artifact Service	PAS	The component that assembles and seals a Provenance Artifact from the correlated ALS evidence set. It is invoked only when both compliance signals are active. Export remains blocked until sealing succeeds. PAS is independent of the execution profile and runs after the evidence required by the approved artifact contract is available.
Registered data source	N/A	A governed warehouse, lakehouse, semantic service, approved data API, relationship store, or multidimensional engine available to the FQE through an approved connection and mapping. Registration requires defined credentials, classifications, timeout behavior, capabilities, and failure semantics.
Result Cache	N/A	An optional cache inside the controlled execution boundary. Its canonical key includes the effective plan, tenant, entitlement projection, pinned definition versions, and relevant configuration. Every cache hit still passes current authorization and creates new lineage evidence. Compliance-purpose requests bypass the cache unless approved policy provides equivalent evidence and freshness guarantees.

Term	Abbreviation	Definition
Role-Aware Projection Layer	RAPL	The deterministic component that combines verified identity context with current DES policy to produce the caller's effective entitlement projection. The projection contains permitted concepts, row scope, masks, classification ceiling, policy versions, and decision evidence.
Semantic Controls Layer	SCL	The mandatory release gate between semantic validation and physical planning. It evaluates data scale, complexity, classification, compliance, and concurrency against versioned controls, assigns an execution timeout, records the decision, and rejects requests that may not proceed. When the compliance trigger is active, it marks the provenance artifact as required and export as blocked pending PAS.
Semantic Data Repository	SDR	The governed source of business and physical data metadata, including data models, critical data elements, quality rules, structures, mappings, profiling evidence, and data-lineage references. It is separate from the SMR , although both may use the DCS . Adoption requires a readiness assessment of its coverage, ownership, freshness, and mapping quality; the architecture does not assume that every organization already has a suitable SDR.
Semantic Metrics Repository	SMR	The governed catalog of approved analytical metrics, dimensions, operations, dataset contracts, presentation metadata, and mapping references. Definitions are owned, versioned, reviewed, and approved before governed resolution. The SMR defines analytical meaning; the SDR describes the underlying data context. Search indexes or embeddings may support discovery but do not change approval state or access rights.

Term	Abbreviation	Definition
Semantic Validation Layer	SVL	The deterministic semantic quality gate that receives a resolved request, entitlement projection, and approved SMR definitions and produces a backend-independent LQP or structured rejection. It validates parameters, pins versions, checks semantic compatibility, applies the entitlement projection, and attaches validated compliance inputs. No language model runs in this component.
Structured Response	N/A	The governed delivery envelope returned to a consumer. It distinguishes status, typed data or a data reference, DVL specification, optional narrative, lineage, compliance state, warnings, and errors. Consumers render this contract without recalculating or reinterpreting governed values.

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